

RPI-T2 with MAGNA3

For heat energy monitoring

Installation and operating instructions



English (GB) Installation and operating instructions

Original installation and operating instructions

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1. General information

This document is intended for the use of the RPI-T2 sensor as heat energy monitor on the MAGNA3 pump. It describes how to connect the sensor and how to configure the pump analog input for heat energy monitoring.

The RPI-T2 can also be used for pressure measurement. This feature is not described in this document.

2. Symbols used in this document



WARNING

Indicates a hazardous situation which, if not avoided, could result in death or serious personal injury.

3. Electrical connection

WARNING

Electric shock

Death or serious personal injury

- Before starting any work on the product, make sure that the power supply has been switched off and that it cannot be accidentally switched on.

1. Remove the front cover of the pump and locate the following terminals.

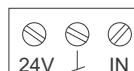


Fig. 1 Analog input for external sensor

2. Connect the sensor wires to the terminals according to the diagram below.

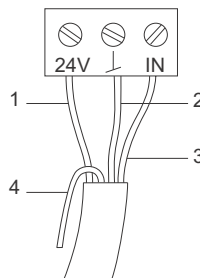


Fig. 2 Wiring diagram for temperature measurement

Pos.	Wire colour	Description
1	Brown	Power supply
2	Blue	GND
3	Black	Temperature signal, 0-10 V
4	White	Pressure signal, 0-10 V

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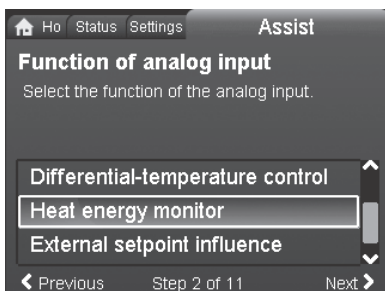
4. Setting the analog input

You can set the analog input for heat energy monitoring either via the MAGNA3 display or via the Grundfos GO app.

The heat energy monitor is a monitoring function that calculates the heat energy consumption within a system.

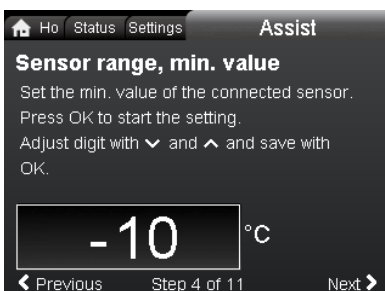
4.1 MAGNA3

1. Navigate to "Assist" > "Setup, analog input".
2. In "Function of analog input", select "Heat energy monitor".



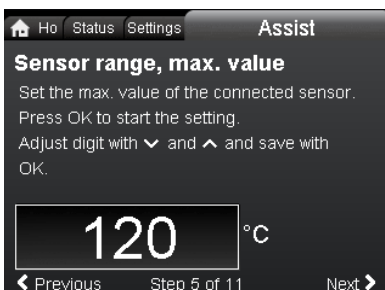
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3. Select temperature unit. Press "Next".
4. Enter the minimum sensor measuring range: -10 °C (14 °F). Press "Next".



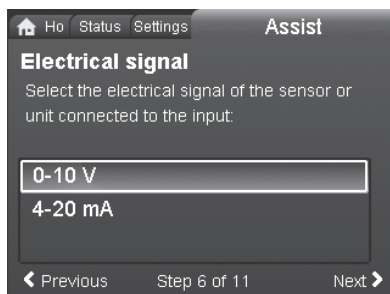
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5. Enter the maximum sensor measuring range: 120 °C (248 °F). Press "Next".



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6. Select the electrical signal: "0-10 V". Press "Next".



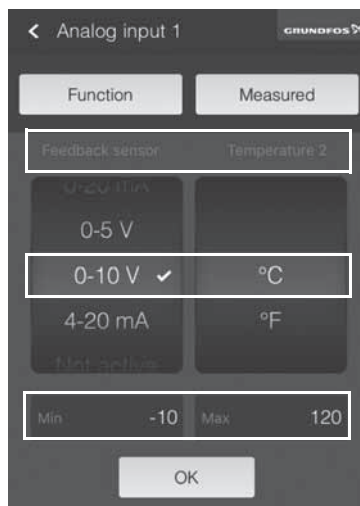
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7. Check the summary of the settings displayed and press [OK].

4.2 Grundfos GO

1. From the main dashboard, navigate to "Settings" > "Analog input 1".
2. In the "Analog input 1" menu, set the sensor as described in the table below and press "OK".

Function	Feedback sensor
Measured value	Temperature 2
Electrical signal	0-10 V
Unit	°C or °F
Minimum measuring range	-10 °C (14 °F)
Maximum measuring range	120 °C (248 °F)



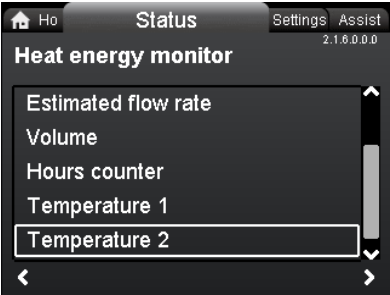
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The function is set, and you can go back to the main menu.

5. Reading the temperature

5.1 MAGNA3

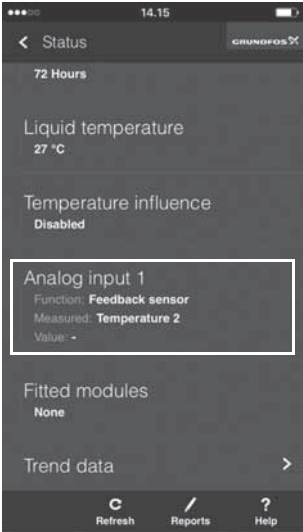
- 1. Navigate to "Status" > "Heat energy monitor" > "Temperature 2".



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5.2 Grundfos GO

- 1. On the main dashboard, press "Status".
- 2. Read the temperature under "Analog input 1".



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6. Technical data

Temperature	
Measuring range	-10 to +120 °C (14-248 °F)
Accuracy (± 1 σ), 0-80 °C (32-176 °F)	± 1 K
Accuracy (± 1 σ), 0-100 °C (32-212 °F)	± 2 K
Response time for sensor electronics	Less than 100 ms (typically 50 ms)
Resolution	0.1 K
System conditions and environment	
Liquid types	Liquids compatible with wetted materials
Liquid temperature, operation	-30 to +120 °C (-22 to +248 °F)
Ambient temperature, operation	-25 to +60 °C (-13 to +140 °F)
Storage temperature	-55 to +70 °C (-67 to +158 °F)
Humidity	0-95 % RH, non- condensing
Maximum system pressure	25 bar (360 psi)
Electrical data	
Power supply, RPI+T	16.6 - 30 VDC
Output signals – Signal cut off	0-10 VDC 11 VDC
Maximum power consumption	300 mW
Minimum load impedance	10 kΩ
Maximum cable length	Less than 30 m (98 ft)
Materials	
Sensing element	Silicon-based MEMS
Sealing	EPDM or FKM
Housing	Stainless steel 1.4404 (AISI 316 L)
Wetted materials	Corrosion-resistant coating EPDM or FKM Stainless steel 1.4404 (AISI 316 L)
Environmental standards	
Enclosure class	IP67, cable connected
Temperature cycling	IEC 68-2-14
Vibration, non-destructive	20-2000 Hz, 10G, 4 h
Electromagnetic compatibility	EN 61326-1

Subject to alterations.

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